LU Decomposition

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What is LU Decomposition?

- LU decomposition of a square matrix A is a factorization of A as the product of a lower triangular matrix L and an upper triangular matrix U.
- In this factorization, we can assume that all the diagonal entries of L are equal to one.
- The LU decomposition of A, if it exists, can be obtained by reducing A into an echelon form.

Why use LU Decomposition?

- The LU decomposition is primarily used to solve linear systems of equations $AX = b$.
- To do this, we first factorize A as $A = LU$.
- Next, we solve the equation $LY = b$ for Y using forward substitution.
- Finally, we can find X from the equation $UX = Y$ by using back substitution.

Problem 1

Solve the following system of equations by LU decomposition method:

$$\begin{aligned} 3x - 7y - 2z &= -7 \\ -3x + 5y + z &= 5 \\ 6x - 4y &= 2 \end{aligned}$$

Step 1: LU decomposition of the coefficient matrix A

The coefficient matrix A is:

$$A = \begin{pmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{pmatrix}$$

Performing elementary row operations to get U and L:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + 1 \cdot R_1} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 6 & -4 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - 2 \cdot R_1} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 10 & 4 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 + (-5) \cdot R_2} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{pmatrix} \begin{pmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{pmatrix}$$

So,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{pmatrix}$$

Step 2: Solve $LY = b$

Let $Y = [y_1, y_2, y_3]^T$ and $b = [-7, 5, 2]^T$. We solve $LY = b$:

$$\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -7 \\ 5 \\ 2 \end{pmatrix}$$

From the first row: $y_1 = -7$. From the second row: $-y_1 + y_2 = 5 \implies -(-7) + y_2 = 5 \implies 7 + y_2 = 5 \implies y_2 = -2$. From the third row: $2y_1 - 5y_2 + y_3 = 2 \implies 2(-7) - 5(-2) + y_3 = 2 \implies -14 + 10 + y_3 = 2 \implies -4 + y_3 = 2 \implies y_3 = 6$. So, $Y = [-7, -2, 6]^T$.

Step 3: Solve $UX = Y$

Let $X = [x_1, x_2, x_3]^T$. We solve $UX = Y$:

$$\begin{pmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -7 \\ -2 \\ 6 \end{pmatrix}$$

From the third row: $-x_3 = 6 \implies x_3 = -6$. From the second row: $-2x_2 - x_3 = -2 \implies -2x_2 - (-6) = -2 \implies -2x_2 + 6 = -2 \implies -2x_2 = -8 \implies x_2 = 4$. From the first row: $3x_1 - 7x_2 - 2x_3 = -7 \implies 3x_1 - 7(4) - 2(-6) = -7 \implies 3x_1 - 28 + 12 = -7 \implies 3x_1 - 16 = -7 \implies 3x_1 = 9 \implies x_1 = 3$. So, $X = [3, 4, -6]^T$. **Solution:** $x_1 = 3, x_2 = 4, x_3 = -6$.

Problem 2

Solve the following system of equations by LU decomposition method:

$$\begin{aligned} 2x - y + 2z &= 1 \\ -6x - 2z &= 0 \\ 8x - y + 5z &= 4 \end{aligned}$$

Step 1: LU decomposition of the coefficient matrix A

The coefficient matrix A is:

$$A = \begin{pmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{pmatrix}$$

Performing elementary row operations to get U and L:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + 3 \cdot R_1} \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 8 & -1 & 5 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - 4 \cdot R_1} \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 3 & -3 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 + 1 \cdot R_2} \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

So,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

Step 2: Solve $LY = b$

Let $Y = [y_1, y_2, y_3]^T$ and $b = [1, 0, 4]^T$. We solve $LY = b$:

$$\begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$$

From the first row: $y_1 = 1$. From the second row:
$-3y_1 + y_2 = 0 \implies -3(1) + y_2 = 0 \implies y_2 = 3$. From the third row:
$4y_1 - y_2 + y_3 = 4 \implies 4(1) - 3 + y_3 = 4 \implies 1 + y_3 = 4 \implies y_3 = 3$.
So, $Y = [1, 3, 3]^T$.

Step 3: Solve $UX = Y$

Let $X = [x_1, x_2, x_3]^T$. We solve $UX = Y$:

$$\begin{pmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}$$

From the third row: $x_3 = 3$. From the second row: $-3x_2 + 4x_3 = 3 \implies -3x_2 + 4(3) = 3 \implies -3x_2 + 12 = 3 \implies -3x_2 = -9 \implies x_2 = 3$. From the first row: $2x_1 - x_2 + 2x_3 = 1 \implies 2x_1 - 3 + 2(3) = 1 \implies 2x_1 - 3 + 6 = 1 \implies 2x_1 + 3 = 1 \implies 2x_1 = -2 \implies x_1 = -1$. So, $X = [-1, 3, 3]^T$. **Solution:** $x_1 = -1, x_2 = 3, x_3 = 3$.

Exercise 1

Find an LU decomposition of the matrix:

$$A = \begin{pmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{pmatrix}$$

Solution

We want to find L and U such that $A = LU$.

$$A = \begin{pmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{pmatrix}$$

Perform row operations to transform A into an upper triangular matrix U. The multipliers used for row operations will form the L matrix.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - (-1)\cdot R_1} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 8 & 6 & -8 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - 2\cdot R_1} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{pmatrix}$$

So,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{pmatrix}$$

Exercise 2

Solve the following system of equations by LU decomposition method:

$$\begin{aligned} 2x - 2y + 4z &= 0 \\ x - 3y + z &= -5 \\ 3x + 7y + 5z &= 7 \end{aligned}$$

Step 1: LU decomposition of the coefficient matrix A

The coefficient matrix A is:

$$A = \begin{pmatrix} 2 & -2 & 4 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{pmatrix}$$

Performing elementary row operations to get U and L:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 & 4 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - (1/2)\cdot R_1} \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 & 4 \\ 0 & -2 & -1 \\ 3 & 7 & 5 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - (3/2)\cdot R_1} \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 3/2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 & 4 \\ 0 & -2 & -1 \\ 0 & 10 & -1 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - (-5)\cdot R_2} \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 3/2 & -5 & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 & 4 \\ 0 & -2 & -1 \\ 0 & 0 & -6 \end{pmatrix}$$

So,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 3/2 & -5 & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 2 & -2 & 4 \\ 0 & -2 & -1 \\ 0 & 0 & -6 \end{pmatrix}$$

Step 2: Solve $LY = b$

Let $Y = [y_1, y_2, y_3]^T$ and $b = [0, -5, 7]^T$. We solve $LY = b$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 3/2 & -5 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -5 \\ 7 \end{pmatrix}$$

From the first row: $y_1 = 0$. From the second row: $(1/2)y_1 + y_2 = -5 \implies (1/2)(0) + y_2 = -5 \implies y_2 = -5$. From the third row: $(3/2)y_1 - 5y_2 + y_3 = 7 \implies (3/2)(0) - 5(-5) + y_3 = 7 \implies 25 + y_3 = 7 \implies y_3 = -18$. So, $Y = [0, -5, -18]^T$.

Step 3: Solve $UX = Y$

Let $X = [x_1, x_2, x_3]^T$. We solve $UX = Y$:

$$\begin{pmatrix} 2 & -2 & 4 \\ 0 & -2 & -1 \\ 0 & 0 & -6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -5 \\ -18 \end{pmatrix}$$

From the third row: $-6x_3 = -18 \implies x_3 = 3$. From the second row: $-2x_2 - x_3 = -5 \implies -2x_2 - 3 = -5 \implies -2x_2 = -2 \implies x_2 = 1$. From the first row: $2x_1 - 2x_2 + 4x_3 = 0 \implies 2x_1 - 2(1) + 4(3) = 0 \implies 2x_1 - 2 + 12 = 0 \implies 2x_1 + 10 = 0 \implies 2x_1 = -10 \implies x_1 = -5$. So, $X = [-5, 1, 3]^T$. **Solution:** $x_1 = -5, x_2 = 1, x_3 = 3$.

Exercise 3

Solve the following system of equations by LU decomposition method:

$$\begin{aligned} 4x + 2y + 14z &= 14 \\ 2x + 17y - 5z &= -101 \\ 14x - 5y + 83z &= 155 \end{aligned}$$

Step 1: LU decomposition of the coefficient matrix A

The coefficient matrix A is:

$$A = \begin{pmatrix} 4 & 2 & 14 \\ 2 & 17 & -5 \\ 14 & -5 & 83 \end{pmatrix}$$

Performing elementary row operations to get U and L:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 2 & 14 \\ 2 & 17 & -5 \\ 14 & -5 & 83 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - (1/2)\cdot R_1} \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 2 & 14 \\ 0 & 16 & -12 \\ 14 & -5 & 83 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - (7/2)\cdot R_1} \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 7/2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 2 & 14 \\ 0 & 16 & -12 \\ 0 & -12 & 34 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - (-3/4)\cdot R_2} \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 7/2 & -3/4 & 1 \end{pmatrix} \begin{pmatrix} 4 & 2 & 14 \\ 0 & 16 & -12 \\ 0 & 0 & 25 \end{pmatrix}$$

So,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 7/2 & -3/4 & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 4 & 2 & 14 \\ 0 & 16 & -12 \\ 0 & 0 & 25 \end{pmatrix}$$

Step 2: Solve $LY = b$

Let $Y = [y_1, y_2, y_3]^T$ and $b = [14, -101, 155]^T$. We solve $LY = b$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 7/2 & -3/4 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 14 \\ -101 \\ 155 \end{pmatrix}$$

From the first row: $y_1 = 14$. From the second row: $(1/2)y_1 + y_2 = -101 \implies (1/2)(14) + y_2 = -101 \implies 7 + y_2 = -101 \implies y_2 = -108$.
From the third row:
$(7/2)y_1 - (3/4)y_2 + y_3 = 155 \implies (7/2)(14) - (3/4)(-108) + y_3 = 155 \implies 49 + 81 + y_3 = 155 \implies 130 + y_3 = 155 \implies y_3 = 25$. So, $Y = [14, -108, 25]^T$.

Step 3: Solve $UX = Y$

Let $X = [x_1, x_2, x_3]^T$. We solve $UX = Y$:

$$\begin{pmatrix} 4 & 2 & 14 \\ 0 & 16 & -12 \\ 0 & 0 & 25 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 14 \\ -108 \\ 25 \end{pmatrix}$$

From the third row: $25x_3 = 25 \implies x_3 = 1$. From the second row: $16x_2 - 12x_3 = -108 \implies 16x_2 - 12(1) = -108 \implies 16x_2 - 12 = -108 \implies 16x_2 = -96 \implies x_2 = -6$. From the first row: $4x_1 + 2x_2 + 14x_3 = 14 \implies 4x_1 + 2(-6) + 14(1) = 14 \implies 4x_1 - 12 + 14 = 14 \implies 4x_1 + 2 = 14 \implies 4x_1 = 12 \implies x_1 = 3$. So, $X = [3, -6, 1]^T$. **Solution:** $x_1 = 3, x_2 = -6, x_3 = 1$.